Check for updates

Regular paper

Predictive analytics for de novo malignancies after lung transplantation

Intelligent Decision Technologies
2025, Vol. 19(2) 1040–1053
© The Author(s) 2024
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/18724981241296492
journals.sagepub.com/home/idt
Sage | IOS Press

Amir Zadeh[1], Nasim Nosoudi[2], Raegan Halley[2], Cameron Kiani[3] and Jaime E Ramirez-Vick[4]

Abstract

Early identification and detection of cancer following lung transplantation is crucial for improving survival rates. This study utilizes data analytics techniques to predict de novo malignancies after lung transplantation in recipients with no prior cancer history. The purpose of this study is to identify critical factors influencing post-transplant malignancy occurrence to improve survival rates. A dataset of 30,917 individuals, 25% of whom developed post-transplant malignancy, was utilized in this study. Several well-known machine learning techniques were applied to this dataset to develop predictive models for malignancy following lung transplantation. The findings indicated that gradient boosting outperformed all other predictive models trained with the dataset, achieving an area under the ROC curve (AUC) of 0.746. The analysis revealed that the recipient's DR52 Antigen, BMI, CMV status, and total serum albumin levels were identified as critical factors influencing the occurrence of de novo malignancies. Lung transplant recipients face heightened de novo post-transplant malignancy risk due to immunosuppression. This study particularly underscores HLA-DR's association with a reduced likelihood of malignancy. Tailored surveillance protocols for lung transplant recipients could help mitigate this risk, if proven cost-effective, ensuring optimal care for this vulnerable patient group.

Keywords

Lung transplantation, de novo malignancy, machine learning, gradient boosting, HLA-DR, DR52 antigen

Received: 9 October 2024; accepted: 16 October 2024

1 Introduction

For people with end-stage organ failure, transplantation is a potentially life-saving intervention. Post-transplant treatments have made remarkable progress over time, enhancing first-year survival rates after transplantation. Despite improvements in solid organ transplantation outcomes, complications, including a heightened risk of malignancy, remain a significant concern. Immunosuppressive drugs, crucial for preventing organ rejection, can also impair the immune system's ability to identify and eliminate cancer cells, thereby increasing the risk of post-transplant malignancy.[1]

Transplant recipients are two to seven times more likely to develop de novo malignancies compared to the general population.[2] The most common type of cancer in people who have had organ transplants is non-melanoma skin cancer (NMSC). Kaposi sarcoma, non-Hodgkin lymphoma, liver, anus, vulva, and lip cancers are other cancers associated with organ transplants.[3] Therefore, following organ transplantation, there is a significant demand for effective methods to monitor and detect post-transplant malignancy.

[1]Department of Information Systems, Raj Soin College of Business, Wright State University, Dayton, OH, USA
[2]Department of Biomedical Engineering, College of Engineering and Computer Sciences, Marshall University, Huntington, WV, USA
[3]Division of Occupational Medicine, Department of Environmental Medicine and Climate Science, Icahn School of Medicine at Mount Sinai, New York, NY, USA
[4]Department of Biomedical Engineering and Human Factor, College of Engineering, Wright State University, Dayton, OH, USA

Corresponding author:
Amir Zadeh, Department of Information Systems, Raj Soin College of Business, Wright State University, 3640 Colonel Glenn Hwy, Dayton, OH 45435, USA.
Email: amir.zadeh@wright.edu

With the recent advancements in explainable artificial intelligence (AI) and machine learning (ML), coupled with the rise of big data,[4] the quest for more effective and accurate algorithms to predict malignancy in transplant recipients has become more promising than ever before.[5] By utilizing extensive datasets from diverse sources such as electronic health records (EHRs) and the United Network for Organ Sharing transplant registry (UNOS-TR), machine learning techniques can identify patterns and make accurate predictions. These algorithms have already been employed to predict diverse outcomes in recipients of solid organ transplants.[6–8]

Relative to traditional risk scores, machine learning algorithms show promising potential in predicting graft and patient survival by utilizing pre-transplant variables.[6–8] Previous studies have mainly concentrated on predicting overall survival rates[9–11] rather than identifying prognostic factors for malignancy. Only a limited number of studies have explored the prediction of de novo malignancy outcomes post-organ transplant, using longitudinal data collected both prior to and after transplantation.[12] However, none of these studies have specifically addressed de novo malignancy following lung transplantation. Therefore, the objective of this research is to employ a machine learning approach to construct a series of predictive models for predicting de novo malignancies in patients who have undergone lung transplantation. Utilizing data obtained from the Standard Transplant Analysis and Research (STAR), we developed multiple predictive models to classify malignancy patterns in organ transplant recipients and identify significant prognostic factors impacting their risk of developing malignancy post-transplantation. The medical records of lung transplant recipients between 1984 and 2021 were utilized for this study. A training set was used for developing the prediction models, while a validation set was created for model validation to ensure robustness and generalizability to unseen data. All algorithms were tested on the STAR database, and the best-performing model for our study was reported. The performance of these models was evaluated using the area under the receiver operating characteristic (ROC) curve, among other metrics.

2 Literature review

Organ transplantation stands as a highly effective treatment for end-stage respiratory failure.[13] With the advent of AI and ML in the medical field, transplant clinicians are increasingly exploring these technologies as decision-support tools in crucial scenarios. These include predicting waiting list mortality, aiding in donor selection, forecasting patient and graft survival rates, predicting de novo malignancies, and optimizing immunosuppression strategies to prevent rejection.[8,12]

ML plays a crucial role in addressing the increasing time patients spend on transplant waiting lists, highlighting concerns about unequal organ access.[8] Informed decisions regarding patient listing, organ allocation, and donor organ acceptance depend on accurate waiting list mortality and morbidity.[14] ML algorithms have excelled in identifying unknown risk factors influencing transplant outcomes by using a blend of factors such as patient demographics, genetics, comorbidities, and graft quality.[14]

Other ML studies have directed their focus toward post-transplant outcomes rather than waitlist mortality, integrating donor and recipient characteristics to enhance transplant success rates. Given the complexity and ethical considerations surrounding organ allocation and donor-recipient matching, testing different allocation systems in real-world experimental settings is impractical. However, ML offers the capability to simulate and explore various scenarios. For instance, Davis et al.[15] utilized multiperiod linear optimization models to evaluate different kidney sharing practices and their outcomes prior to implementation. Likewise, other studies[16,17] have applied machine learning algorithms such as artificial neural networks (ANN) to both donor and recipient data to examine a range of donor-recipient matching methods for liver transplantation.

In addition, given the scarcity of organs relative to the demand from patients, predicting the outcomes of organ transplantation becomes crucial for optimizing patient care and allocating resources effectively.[8] Various studies have proposed ML algorithms to identify predictive factors impacting patient survival after organ transplants.[18–21] These models identified various factors such as graft failure, infections, renal failure, and vascular complications. ML has also been successfully applied to predict short-term postoperative survival for organ transplantation based on preoperative physiological measurements. These models leveraged patients' blood test data within 1–9 days before surgery.[22] Similarly, Lau et al.[23] devised a personalized predictive model for 90-day graft failure in pediatric liver transplantation using machine learning. This model highlighted features like preoperative hepatic encephalopathy, sodium level at the end of surgery, hepatic artery thrombosis, and total bilirubin level as significant predictors.

In thoracic transplantation, Amini et al.[24] introduced an analytical framework aimed at identifying crucial factors influencing long-term survival following lung transplantation. Analyzing data from over 44 thousand lung transplant patients, the proposed framework underscored factors such as hepatitis B surface antibody levels and forced expiratory volume in one second (FEV1) as significant contributors. These insights offer valuable guidance for medical scholars and practitioners seeking to optimize the lung allocation system. Similarly, Medved et al.[25] conducted a study comparing a deep-learning-based prediction model with a traditional risk-based model (i.e., logistic regression) to forecast short-

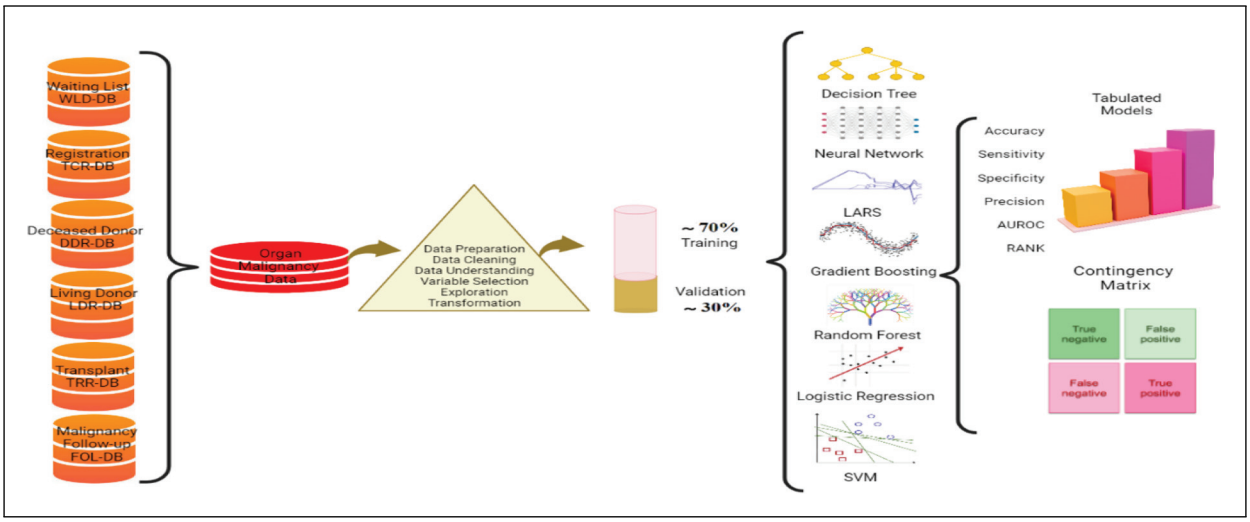

Figure 1. Research framework.

and long-term survival after heart transplantation. Their findings revealed that the deep-learning model exhibited superior performance, surpassing the traditional approach in accurately predicting survival outcomes.

Despite all these studies addressing various issues surrounding organ transplantation, there has been limited focus on assessing the risk of de novo malignancy following organ transplantation.[12,26] Notably, none of these studies have specifically investigated de novo malignancies following lung transplantation. Our study, therefore, aims to fill this gap by developing ML models specifically tailored to predict malignancy in patients who have undergone lung transplantation, utilizing longitudinal pre- and post-transplant data.

3 Materials and methods

Our research methodology comprises three fundamental elements: data acquisition, data preparation, and data modeling.[27] Each component will be explained in detail in the following subsections, accompanied by a comprehensive visualization in Figure 1. The dataset used in this study was obtained from the Standard Transplant Analysis and Research (STAR) data sources. STAR, overseen by the United Network for Organ Sharing (UNOS), plays a vital role in supporting the procurement and operations of the Organ Procurement and Transplantation Network (OPTN) in the United States. These sources contain detailed individual-level information for all transplants conducted in the U.S. since 1987 up to the present day, making it the most extensive organ transplant patient database in the country. To prepare the data for predictive modeling, we performed data cleaning and processing to transform it into a suitable format. SAS Enterprise Miner was employed for data analysis and exploration.

3.1 Data acquisitions

In our study, we created a dataset that incorporated information sourced from various databases, as depicted in Figure 1. The transplant waitlist data source offered valuable insights into each patient's demographic characteristics, PRA sensitivity, HLA antigen levels, and other significant factors. Additionally, the transplant data sources included procedural data from the transplant operations. The living and deceased donor databases contributed supplementary information on donor serology, organ recovery and preservation, expected organ beneficiaries, and donor demographics. Finally, data pertaining to patients diagnosed with post-transplant malignancy was extracted from the lung malignancy follow-up data source. The initial dataset for this study contained 196,270 patient records with theoretic transplants. Table 1 shows the number of theoretical transplantations that occurred in the United States. It is important to note that for the purpose of this study, our focus was exclusively on lung transplantation. Therefore, heart and heart-lung transplantations were not included in our analysis.

Based on the collected data, the key variables can be classified into three distinct categories: 1) pre-transplant measures encompassing recipient and donor demographic information such as gender, race, and age; 2) during-transplant measures, consisting of blood work results; and 3) post-transplant measures including patient outcomes such as survival, hospital

Table 1. Theoretic transplantation statistics.

Organ	STAR code	Frequency
Lung	LU	66, 590
Heart	HR	125, 794
Heart-Lung	HL	3462

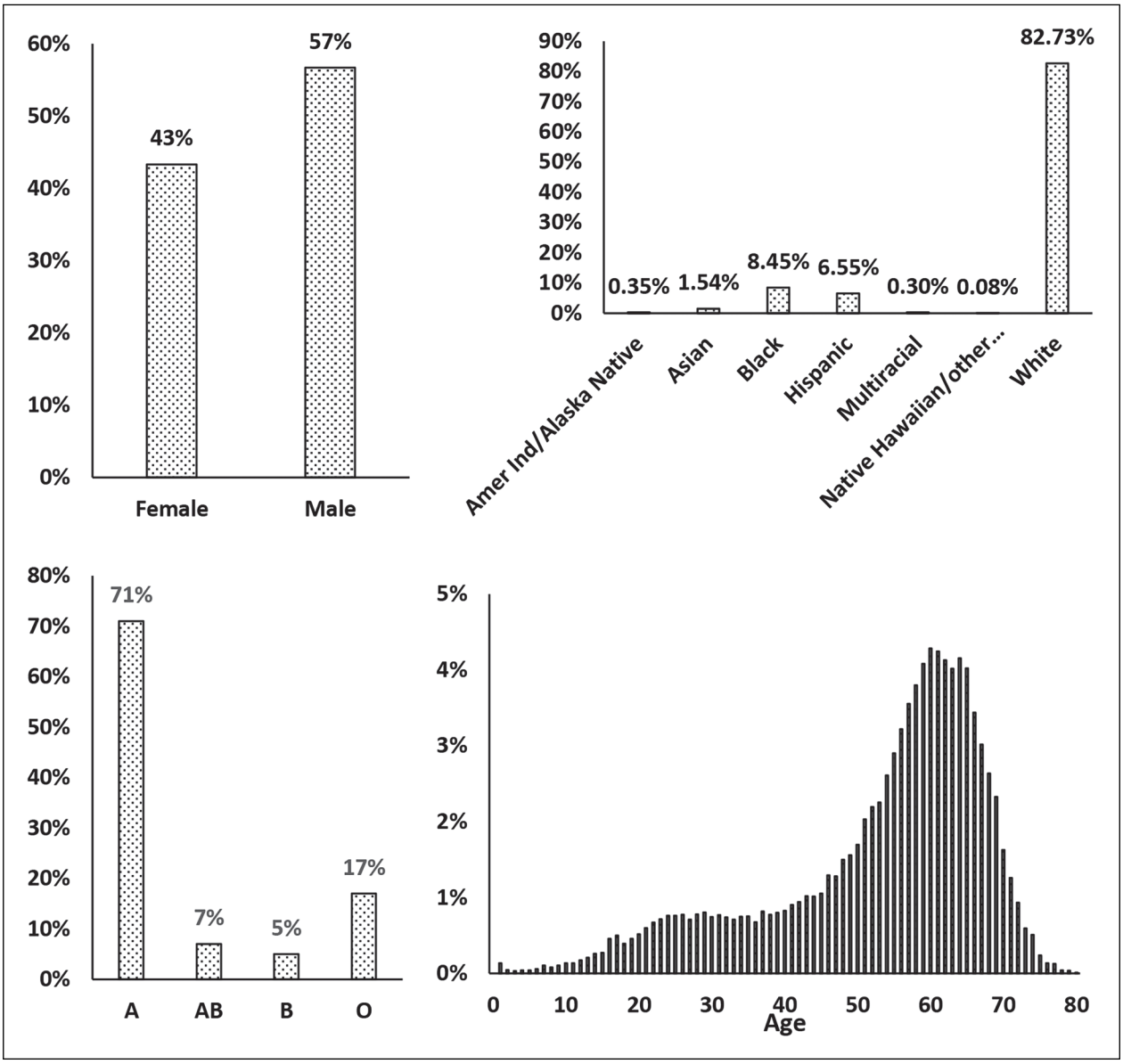

Figure 2. Demographic snapshot: gender, ethnicity, blood type, and age distribution.

length of stay, presence of malignancy, and other complications. Figure 2 illustrates the demographic distribution within the dataset utilized in this study.

3.2 Data preparation

After consolidating data from multiple sources for this research, extensive time and effort were dedicated to preprocessing, cleaning, and understanding the data.[28] To focus specifically on de novo malignancy following lung transplantation, we extracted records exclusively related to lung transplants from the original database. This resulted in a total of 66,590

cases representing patients who underwent lung transplantation. Subsequently, individuals with a documented history of malignant cancer, as indicated in their transplant candidate forms, were removed from the cohort. A total of 38,788 cases with no prior history of malignancy remained. By merging this dataset with follow-up data on lung malignancy, we identified individuals who were diagnosed with malignant cancers during their respective follow-up periods. After excluding those who were lost to follow-up, our final data table consisted of 30,917 individuals, among whom 25% developed malignancy after lung transplantation.

Following the merging process, the dataset encompassed a total of 544 variables. However, to streamline the analysis and prioritize the most relevant variables, medical experts with expertise in lung transplantation procedures carefully evaluated each variable. Their assessment aimed to identify potential issues such as irrelevance, redundancy, and a high percentage of unacceptable missing values (>30%). Moreover, the strength of the relationship between each input variable and the target variable was assessed to identify excessively correlated variables that could negatively impact the predictive model's performance. As a result of this evaluation, a total of 173 variables were excluded from the study.

3.3 Data modelling

This study conducted a supervised investigation focused on a set of variables to gain deeper insights into the occurrence of malignancy in patients who have undergone solid organ transplantation. To achieve this objective, three distinct classes of predictive models were developed: decision trees, logistic regression, and neural networks. The decision tree models included entropy, boosting trees, Gini, chi-square, and random forests. Logistic regression modeling encompassed stepwise, backward, forward, and least angle regression techniques. Each model, whether decision tree, logistic regression, or neural network, was developed according to industry standards to ensure a robust approach for empirical research and the examination of malignancy patterns. To evaluate the performance of the models, the dataset was divided into two parts: 70% for training and 30% for testing. Given the presence of data imbalance, oversampling was employed to increase the percentage of the minority class by replicating instances from the minority class.

3.3.1 Logistic regression. Logistic regression is a widely used machine learning method that aims to predict the probability of an observation belonging to a certain class based on one or more input variables. In logistic regression, the target variable takes on binary values (0 or 1). Given a column vector of input variables denoted as X, the logistic regression model estimates the probability, denoted as $p = \Pr(Y = 1|X)$, where Y represents the target variable. The logistic regression model can be represented mathematically as follows:

$$logit(\pi) = \log\left(\frac{p}{1-p}\right) = \alpha + \beta^T X \tag{1}$$

The intercept parameter in logistic regression is represented by α, while the vector of slope parameters is denoted as β^T. The estimation of these parameters is typically done using the maximum log-likelihood estimation (MLE) method. MLE involves maximizing the logarithm of the likelihood function, which measures the likelihood of observing the given data under the assumed logistic regression model. By maximizing the log-likelihood function, we can obtain estimates for the intercept and slope parameters that best fit the data.[29]

In this study, logistic regression models were constructed with the aim of achieving the highest possible accuracy in predicting the target variable (Y), which represents the presence or absence of malignant cancer. When an individual is diagnosed with malignant cancers, Y is assigned a value of 1, whereas a value of 0 indicates the absence of malignancy. The variable selection methods employed in the model-building process included forward selection, backward elimination, and stepwise selection. All of these methods considered all available predictors in order to identify the most influential variables for predicting the target variable.[30]

Forward selection is an iterative variable selection method that initially starts with an empty model. It gradually adds one variable at a time, continuing until no additional variables meet a predefined significance level (such as a p-value < 0.05). On the other hand, backward selection begins with all attributes included in the model and sequentially eliminates the least significant variable until only variables with p-values lower than a predetermined significance level (e.g., < 0.05) remain in the model. Lastly, stepwise selection combines elements of both forward and backward selection. At each step, it adds an input variable, and any variable that becomes nonsignificant is removed from the model. This process is repeated until none of the discarded variables have p-values lower than the predetermined significance level for entry, and all variables in the model are significant based on a predefined significance level for staying in the model.[31] In this study, logistic regression models were constructed using forward, backward, and stepwise variable selection methods. The stepwise selection method was chosen as it yielded the most optimal solution for this particular study.

3.3.2 Least angle regressions (LARS). The LARS (Least Angle Regression) model differs from a standard regression model by considering only predictors that meet predefined criteria based on Schwartz's Bayesian Criterion (SBC), Akaike's Information Criterion (AIC), or adjusted R^2. It effectively handles multicollinearity and high-dimensional data by selecting the most relevant predictors. The LARS algorithm determines the optimal model trajectory using the angles between predictor variables. It starts with all β coefficients set to zero and no variables, then identifies the input variable X_j with the highest correlation to Y and adjusts the corresponding coefficient β_j accordingly. The adjustment continues by considering the collective least squares direction of coefficients β_j and β_k until a different input variable with a more significant correlation to the residuals is found. This process is repeated until all input variables are included in the model. In this study, the SBC was chosen as the model selection criterion to achieve the most optimal LARS model. The LARS algorithm's ability to accurately identify predictors and handle multicollinearity has made it a standard machine learning algorithm.[32,33]

3.3.3 Decision tree. Decision trees are highly regarded for their ability to handle missing data and produce easily interpretable results. They employ a rule-based approach that uses input data values to accurately label the dataset. This process involves recursively splitting cases into branches, constructing a tree, and progressively improving classification accuracy.[34] By employing alternative search criteria, significant variables can be identified, and subsequent thresholds are established for further splits. This iterative process continues at each leaf node until the entire tree is formed.[35] In this study, various versions of decision trees are employed to identify the optimal leaf-node splitting criteria. These include the Chi-square log-worth algorithm, which assesses variable significance by measuring the deviation between observed and expected frequencies using the Pearson Chi-square statistic, the Gini algorithm, which utilizes the Gini index for reduction, and the Entropy algorithm, which employs an entropy-based information gain measure for reduction. Finally, pruning algorithms are applied to reduce the complexity of the tree.

The Chi-square log-worth method begins by constructing a two-by-two contingency table, where the columns represent the left and right branches of a decision tree, and each row corresponds to a binary target value (zero or one). The Chi-square test statistic is then calculated based on the frequencies of zeros and ones in the contingency table, comparing the distribution between the left and right branches. The p-value of the Chi-square statistic is then used in the formula shown below to assess the significance of each split:

$$logworth = -log(p - value) \tag{2}$$

This process is iteratively applied to each input variable. Ultimately, the chosen splitting variable will be the one with the highest log-worth value. The Gini and Entropy algorithms determine the worth of a partition or split for a given sample S of classified cases using the following formulas:

$$Gini(S) = 1 - \sum_{i=1}^{c} {p_i}^2 \tag{3}$$

$$Entropy(S) = -\sum_{i=1}^{c} p_i \log_2 p_i \tag{4}$$

The variable p_i represents the fraction of instances in the dataset S that belong to class i. The Gini-based index algorithm measures the purity level achieved by each potential split and identifies the input variable associated with the lowest Gini index value to divide the data.[36] On the other hand, the entropy-based search method evaluates the information gain for each potential attribute and selects the one that yields the highest information gain for dividing the data. This process is repeated for all remaining subsets until the construction of the decision tree is finished.[37] In this study, the decision tree model utilizing the Chi-square method demonstrated superior performance in terms of accuracy, precision, and AUROC rates.

3.3.4 Gradient boosting. The gradient boosting modeling method builds a sequence of decision tree models, with each tree targeting the residual error left by the previous model. This iterative process aims to progressively reduce the overall error, resulting in improved accuracy.[38,39] With the initial weaker tree constructed, the subsequent trees in gradient boosting are fitted to the residuals remaining from the predictions of the prior trees, thereby reducing the prediction error. The procedural steps of the gradient boosting approach can be summarized as follows:

$$F(x) = G_0 + \sum_{i=1}^{n} \beta_i T_i(x) \tag{5}$$

In the gradient boosting approach, the initial value in the series is denoted as G_0. The trees used to predict the residuals are represented by T_i, and the estimated coefficients produced by the gradient boosting algorithm for each respective tree node are denoted as β_i.[36] In this research, the gradient boosting model was utilized to identify important variables in the dataset that were not highlighted by other decision tree algorithms. Before implementing the model, several crucial parameters needed to be adjusted: 1) the number of trees to be constructed (total iterations), 2) the proportion of observations allocated for training each tree, and 3) the shrinkage factor, which controls the learning rate of the model to prevent overfitting.[40] The shrinkage factor, which controls the impact of each individual tree's prediction in the gradient boosting model. In this study, a shrinkage factor of 0.1 was employed to mitigate the risk of overfitting resulting from a large number of iterations. Additionally, the gradient boosting model was configured with 50 iterations and trained using 60% of the available observations.

3.3.5 Random forest. The random forest modeling method combines the predictions of multiple decision trees to produce reliable estimates of the target variable.[41] To build a robust and optimal random forest model, a bootstrap technique is employed to randomly generate samples from the training dataset.[42] Individual trees are constructed using random samples without undergoing a pruning phase to reduce their complexity. Subsequently, each tree in the random forest is used to score validation data based on its respective class labels. The random forest algorithm then determines the predicted class label by considering the votes from each tree in the forest and assigning the class label with the highest vote count as the final prediction.[43] Similar to the gradient boosting model, three parameters need to be adjusted when constructing a random forest: 1) the maximum number of trees to be built, 2) the proportion of training data to be used, and 3) the total number of input variables to include while splitting nodes during the tree-growing process. These parameter choices are crucial for achieving an optimal and effective random forest model.[36] Random forest models are known for their superior performance in handling multi-dimensionality and missing data compared to standard decision trees and other less sophisticated models. In this study, the random forest model was constructed with a maximum of 50 trees, utilizing 60% of the observations as training data. Furthermore, the model was designed to consider no more than 30 input variables when splitting nodes during the tree-growing process. These parameter choices were made to ensure the effectiveness and accuracy of the random forest model in the context of the study.

3.3.6 Neural networks. Artificial neural networks, which fall under the category of machine learning algorithms, possess the capability to capture highly nonlinear relationships present in the variables of a dataset. These models are renowned for their resemblance to the intricate functioning of the human brain.[44]

The Multilayer Perceptron (MLP) is a widely used feed-forward neural network architecture that is known for its ability to effectively model nonlinear relationships. The MLP consists of three layers: the input layer, the hidden layer, and the output layer. The input layer contains neurons corresponding to each input variable, while the hidden layer contains multiple unspecified neurons. The output layer consists of a single neuron representing the target variable. In the MLP, each neuron analyzes its inputs to produce an output, which is then transmitted to the next layer. During the training process, weights are assigned to the connections between neurons in different layers. A weighted sum function is used to combine the inputs and transmit them to a hidden neuron. The inputs x_j and bias terms $b_i^{(1)}$ of each neuron are used to calculate the weighted sum, which is then passed through a hyperbolic tangent or logistic activation function $f^{(1)}$, producing the output of the hidden neuron (denoted as i). The calculation is formulated as follows:

$$h_i = f^{(1)}\left(b_i^{(1)} + \sum_{j=1}^{M} w_{ij}x_j\right) \tag{6}$$

In the MLP, the connecting weights between input j and hidden neuron i are represented by w_{ij}. To calculate class predictions, a logistic activation function $f^{(2)}$ is applied to the output space. The number of hidden neurons is denoted as n_h, and the weights between hidden neuron j and the output are represented by u_j . The formulation for these calculations is as follows:

$$p = f^{(2)}\left(b^{(2)} + \sum_{j=1}^{n_h} u_j h_j\right) \tag{7}$$

The weights of the neural network are initially assigned random values and then adjusted iteratively using the conjugate-gradient optimization method. This optimization method aims to minimize an error function during the training process, allowing the neural network to learn and improve its performance.[36]

3.3.7 Rule induction. Rule induction is a machine learning technique that extracts decision rules from data, providing simple and often highly predictive rules. One common method to generate rule sets is by using a decision tree and converting it into a set of equivalent rules. This approach, known as decision tree rule induction, follows the framework outlined by Quinlan[45] for creating rule sets from trees, as demonstrated by the C4.5 algorithm. Despite their apparent similarities, rules and decision trees differ in terms of the information they extract from datasets and their behavior when applied to new data instances. Decision trees provide a hierarchical structure with detailed attribute tests, while rule induction produces concise if-then rules that are more interpretable.

3.3.8 Support vector machine. Support Vector Machine (SVM) is a powerful class of machine learning algorithms that enables the fitting of a discriminant function using features like polynomial and sigmoid nonlinear kernels to effectively separate different classes. By employing a nonlinear kernel, the input data is transformed into a higher-dimensional space, allowing for the creation of two parallel separating hyperplanes. The main objective is to maximize the distance between these hyperplanes, leading to the development of an optimized classification model. Besides its classification capabilities, SVM can also be utilized for outlier detection and regression tasks. Notably, SVM offers several advantages, including its ability to handle imbalanced data effectively and its relatively lower complexity compared to other classifiers, such as Artificial Neural Networks.[46]

3.3.9 Measures for performance evaluation. There are four common performance metrics used in binary classification problems. The first metric is accuracy, which calculates the overall classification performance of the model. The second metric is the misclassification rate, which is the ratio of the total number of false positive and false negative cases to the total number of cases in the dataset. The third metric is sensitivity, which measures the proportion of actual positives that are correctly classified. The fourth metric is specificity, which measures the proportion of actual negatives that are correctly identified. These performance metrics can be calculated using the following formulas:

$$Accuracy = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}$$
$$Sensitivity = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$
$$Specificity = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}$$

Where TP represents the count of True Positives, TN represents the count of True Negatives, FP represents the count of False Positives, and FN represents the count of False Negatives. These metrics are used to evaluate the performance of prediction models. The accuracy metric provides an overall measure of the model's performance. In the context of detecting malignancy in patients with organ transplants, the sensitivity metric was specifically used to assess the ability of each model to correctly identify positive cases.

4 Results

Table 1 provides a summary of the model performances, including the misclassification rate, accuracy rate, sensitivity rate, and specificity rate for each classifier. The decision tree achieved the highest overall classification rate, while the random forest model exhibited the highest sensitivity compared to the other models. Furthermore, the models were evaluated using the receiver operating characteristic (ROC) index. The ROC curve was plotted to assess the performance of the predictive models in terms of sensitivity and specificity. As depicted in Table 2, the ROC curve confirms that the gradient boosting model outperforms the other models, achieving a ROC value of 0.74.

Table 3 includes the most significant variables for predicting post-transplant malignancy, as determined by gradient boosting. The assigned numbers in Table 3 represent the relative importance of each variable, where a higher value indicates a greater level of significance.

Several interesting findings can be derived from the preceding table. DR52 or Candidate Most Recent/at Removal DR52 Antigen is the most important variable found in this study. Human leukocyte antigens (HLAs) can be found on a person's cells. HLA-DR52 is an HLA-DR serotype that recognizes gene products of HLA-DRB3 locus. Usually, HLA-DR51/52/53 typing is recommended for donors and recipients in kidney transplant.[47] We found that DR52 is associated with decreased probability of de novo malignancy.

END_BMI_CALC is another important factor that we have identified. Although BMI alone does not directly cause cancer, it can be linked to other risk factors that may have an impact on how the disease develops.[48] Chronic inflammation,

Table 2. Classification results.

Method	Accuracy	Sensitivity	Specificity	Precision	AUROC	RANK
Gradient Boosting	0.690	0.731	0.649	0.676	0.746	1
Random Forest	0.663	0.726	0.600	0.645	0.731	2
Decision Tree	0.681	0.713	0.650	0.671	0.729	3
LARS	0.660	0.707	0.613	0.646	0.715	4
Neural Network	0.666	0.716	0.616	0.651	0.712	5
Logistics Regression	0.663	0.727	0.600	0.645	0.711	6
Rule Induction	0.676	0.667	0.685	0.679	0.709	7
SVM	0.644	0.728	0.560	0.623	0.694	8

Table 3. Variable importance of the models.

Variable	Description	From	Validation importance	Rank
DR52	Candidate Most Recent/at Removal DR52 Antigen	WAITING LIST DATA	1.34	1
END_BMI_CALC	Calculated Candidate BMI	CALCULATED	1.33	2
CMV_IGG	Person was infected with CMV	TRR	1.29	3
TOT_SERUM_ALBUM	Recipient total serum albumin at registration	TCR	1.16	4
ECMO_72HOURS	Most recent Recipient's international normalized ratio (INR)	TRR	1.14	5
EBV_SEROSTATUS	Serological status of an individual regarding Epstein-Barr virus (EBV) infection	CALCULATED	1.07	6
LOS	Recipient length of stay	TRR-CALCULATED	1.03	7
TRTREJ1Y	Calculated-treated for rejection within 1 year	CALCULATED	0.97	8
ETHCAT	Recipient ethnicity category	TCR	0.87	9
GENDER	Recipient gender	TCR	0.82	10
INIT_AGE	Age in years at time of listing	CALCULATED	0.80	11
BUN_DON	Deceased donor-terminal blood urea nitrogen	CALCULATED	0.68	12

hormonal imbalances (including elevated estrogen levels), insulin resistance, and weakened immune response are all associated with obesity and can all contribute to the development of cancer.[49]

CMV_IGG is the third most important factor in predicting de novo malignancy following lung transplantation. CMV_IGG describes whether or not a person has cytomegalovirus (CMV)-specific Immunoglobulin G (IgG) antibodies in their blood. The herpesvirus family includes several common viruses, including the cytomegalovirus (CMV). It is pervasive and can infect people of all ages anywhere in the world. CMV can either remain dormant (latent) in the body without creating symptoms or it can induce a wide variety of symptoms, from mild to severe. People with compromised immune systems, such as organ transplant patients or those with HIV/AIDS, are particularly vulnerable to the negative effects of CMV.[50] These individuals may experience unchecked viral replication and ongoing inflammation because of CMV infection, which may raise their risk of developing certain cancers.

TOT_SERUM_ALBUM or recipient total serum albumin at registration is another variable that is significantly associated with de novo malignancy after lung transplant. The relationship between high serum albumin and malignancy is intricate and influenced by multiple factors. Previous research on the link between serum albumin levels and cancer risk has yielded inconsistent results.[51,52] The precise mechanisms behind the association between high albumin levels and the development of malignancy remain incompletely understood. However, the findings of this study suggest a correlation between elevated albumin levels and an increased likelihood of malignancy following lung transplantation.

Extracorporeal Membrane Oxygenation (ECMO) in lung transplant recipients is another significant factor associated with de novo malignancy. The ECMO circuit comprises a typical centrifugal pump and a hollow-fiber oxygenator, which are utilized in cases of severe hypoxic respiratory failure following transplant.[53] Although no direct association between ECMO and malignancy has been established, it is worth considering the potential relationship between the two due to the utilization of ECMO in hypoxic conditions. Further research is needed to explore the potential links between ECMO and the development of malignancy in lung transplant recipients.

The recipient's Epstein-Barr Virus (EBV) is another important factor that exhibited a significant association with de novo malignancy after lung transplant. In other studies, EBV at the time of transplantation was highly associated with

the development of post-transplant lymphoproliferative disorder (PTLD). PTLD is a group of lymphoid disorders that can be attributed to several different types of lymphoma formation, including Hodgkin and non-Hodgkin lymphomas.[54] The suppression of cytotoxic T cells that are responsible for controlling EBV growth due to the immunosuppressive drugs that are required for transplant allows for the uncontrolled growth of the virus. Over 80% of the PTLD cases in the United States and Europe are associated with EBV infection. Lung transplant recipients are shown to have one of the highest incidence rates of PTLD compared to that of other solid organ transplant recipients. Recipients that are EBV seronegative and those who have not been previously exposed to EBV are at an even greater risk of developing malignancy or other complications, especially if the donor has had a history of EBV infection.[55]

Recipient length of stay (LOS) at the hospital has also been identified as an important factor. The length of stay refers to the amount of time a patient spends in the hospital following a transplant or other medical treatment. Although the duration of the patient's stay is not a direct cause of cancer on its own, an extended hospital stay may be associated with certain risk factors, such as an increased likelihood of acquiring infections from pathogens like human papillomavirus (HPV) or hepatitis B and C viruses, which are known to be linked to a higher risk of specific malignancies.[56]

Our findings indicate that "Trtrej1y," which represents being treated for rejection within one year after lung transplantation, is another significant factor in predicting malignancy. The common approach to managing rejection after a transplant involves modifying the patient's immunosuppressive medication regimen. This may entail increasing the dosage of existing medications or introducing new immunosuppressants to enhance immune suppression and reduce the risk of rejection. The specific medications and dosages are tailored to the individual patient's needs and may involve a combination of drugs.[57] Elevating the dosage of immunosuppressive medications may increase the patient's susceptibility to developing cancer. While these medications are crucial for suppressing the immune system to prevent organ rejection, they also diminish the immune system's ability to combat cancer cells, thereby raising the risk of developing malignancies.[58]

Among other variables, gender appeared to be associated with malignancy. It was found that being female has lower risks of malignancy after lung transplant. The male gender is shown to be at a greater risk of developing both melanoma and nonmelanoma skin cancers as well as lung cancer after transplantation.[59]

Another important variable is age which is typically correlated with malignancy after transplantation. As individuals age, they tend to become more susceptible to developing chronic diseases that can contribute to the onset of cancer. This heightened risk is often influenced by a combination of factors including avoidable exposures, modifiable risk behaviors, and the cumulative effects of aging itself.[60]

Lastly, BUN_DON, which represents the blood urea nitrogen levels of the deceased donor, emerged as another important variable in our study. Specifically, for kidney transplantation, one study discovered that higher BUN_DON levels were linked to reduced graft and patient survival time.[61] Other studies have shown a correlation between blood urea nitrogen levels and the prevalence of cancer. This correlation was particularly notable in renal cell carcinomas and breast cancer.[62,63] Hence, our findings provide further confirmation regarding the potential significance of blood urea nitrogen levels of the deceased donor as a valuable predictor of post-transplant malignancy.

5 Discussion

The success of organ transplantation depends on histocompatibility between donor and recipient. The polymorphic human leukocyte antigen (HLA) system plays a critical role in allorecognition between donors and recipients. Specifically, HLA class I and II molecules present antigenic peptides to recipient T cells. Mismatched donor HLA elicits an alloimmune response leading to graft rejection. Therefore, HLA matching is paramount for preventing hyperacute and acute rejection, although immunosuppressive therapies help circumvent HLA incompatibility. While the HLA system is universal, HLA-DR alleles vary among individuals, conferring differences in alloantigen presentation. Specific HLA-DR alleles may increase or decrease cancer risk, although few studies have examined DR52 alleles specifically.

Genes are pieces of DNA that code for proteins that affect physiology and contribute to specific traits or functions. Alleles are many gene variants that differ depending on the nucleotide base found at a specific genomic site. A locus is a distinct, fixed location on a chromosome where a certain gene or genetic marker is located. A haplotype is a collection of alleles from various loci that are shared via transmission on a single chromosome.[64] The human leukocyte antigen (HLA) system is a group of genes that resides on chromosome 6 and encodes proteins that regulate immune responses. These proteins are involved in presenting antigens (foreign substances) to immune cells, which helps initiate and regulate immune responses.[65]

HLA-DR is one of the genes within this system, divided into five major antigen binding groups: DR1, DR51, DR52, DR53 and DR8.[66] While the HLA system is universal and present in all individuals, HLA-DR alleles vary among individuals, conferring differences in alloantigen presentation. Although few studies have examined DR52 alleles specifically,

HLA-DR alleles in general have been found to be associated with an increased risk or decreased risk of developing specific types of cancer.[67]

A 2009 study found that individuals with DR1 and DR52 haplotypes might possess a survival advantage post-transplant. In other words, the presence of these specific haplotypes may be associated with a higher likelihood of surviving after receiving an organ.[68] In another study, DR52 had a protective role against cervical cancer.[69] Specific alleles may be involved in presenting tumor-specific antigens to the immune system, thereby enhancing the immune response against cancer cells. It's also important to note that some HLA-DR alleles have been linked to enhanced immunotherapy responses, a promising treatment option for several malignancies. In some cases, a patient's HLA-DR status is considered when determining their eligibility for a specific immunotherapy.[70]

In conclusion, this study has shown that by analyzing organ recipient data before, during and after transplantation using machine learning algorithms, significant transplant outcomes such as cancer, malignancy, or graft failure can be detected. The findings from this research offer valuable insights for generating personalized recommendations, allowing healthcare providers to identify patients who require specialized clinical attention. Furthermore, this study has confirmed the importance of monitoring variables such as the patient's DR52 antigen, BMI, CMV_IGG, and total serum albumin in individuals who have undergone a lung transplant.

The guidelines for immunosuppression in lung transplant recipients have proposed screening recommendations and immunosuppression protocols for high-risk groups, based on population-level predictions made by machine learning algorithms similar to the one proposed in this research. These algorithms analyze longitudinal data before and after transplantation to assess major outcomes such as malignancy, cancer, infection, or graft failure. Implementing these protocols can help healthcare providers identify individuals who require specialized clinical care. Similar guidelines can be considered for routine screening of HLA-DR following lung transplantation. Future research can focus on evaluating the effectiveness and cost-effectiveness of preventive care measures for transplant recipients at risk of developing malignancies.

6 Conclusion

Lung transplant recipients face an increased risk of developing de novo post-transplant malignancies. Existing literature identifies that immunosuppression is a significant risk factor for the development of these malignancies. The findings of this research emphasize the importance of clinicians recognizing the association between HLA-DR and lower probability of de novo post-transplant malignancies. To mitigate this risk, surveillance protocols can be established specifically for lung transplant recipients. If determined to be cost-effective, these protocols should be implemented as a standard of care for this vulnerable patient population.

Acknowledgments

Raegan Halley was supported in the summer of 2023 by the WV Higher Education Policy Commission, Division of Science and Research under award dsr.20.16 (Summer Undergraduate Research Experiences program).

ORCID iDs

Amir Zadeh https://orcid.org/0000-0002-3171-5629
Nasim Nosoudi https://orcid.org/0000-0003-4286-343X
Jaime E Ramirez-Vick https://orcid.org/0000-0001-9845-3734

Statements and declarations

Authors' contributions

Amir Zadeh: Methodology, Software, Validation, Formal analysis, Data curation, Resources, Writing - original draft. Nasim Nosoudi: Conceptualization, Methodology, Validation, Investigation, Data curation, Writing - original draft. Raegan Halley: Investigation, Data curation, Writing - original draft. Cameron Kiani, Jamie Ramirez-Vick: Validation, Resources, Editing - original draft.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. Gaglio PJ and Cotler SJ. Liver transplantation in adults: long-term management of transplant recipients. 2018. UpToDate [online]. Available at: www. UpToDateInc. com (accessed January 7, 2020).
2. Zwart ES, Yüksel E, Pannekoek A, et al. De novo carcinoma after solid organ transplantation to give insight into carcinogenesis in general—a systematic review and meta-analysis. *Cancers* 2021; 13: 1122.
3. Gogna S, Ramakrishna K and John S. Post transplantation cancer. 2019.
4. Zadeh A, Taylor D, Bertsos M, et al. Predicting sports injuries with wearable technology and data analysis. *Inf Syst Front* 2021; 23: 1023–1037.
5. Massie AB, Kuricka L and Segev DL. Big data in organ transplantation: registries and administrative claims. *Am J Transplant* 2014; 14: 1723–1730.
6. Agasthi P, Buras MR, Smith SD, et al. Machine learning helps predict long-term mortality and graft failure in patients undergoing heart transplant. *General Thorac Cardiovasc Surg* 2020; 68: 1369–1376.
7. Gotlieb N, Azhie A, Sharma D, et al. The promise of machine learning applications in solid organ transplantation. *NPJ Digit Med* 2022; 5: 89.
8. Mekov E and Ilieva V. Machine learning in lung transplantation: where are we? *La Presse Méd* 2022; 51: 104140.
9. Oztekin A, Delen D and Kong ZJ. Predicting the graft survival for heart–lung transplantation patients: an integrated data mining methodology. *Int J Med Inf* 2009; 78: e84–e96.
10. Nursetyo AA, Syed-Abdul S, Uddin M, et al. Graft rejection prediction following kidney transplantation using machine learning techniques: a systematic review and meta-analysis. *MedInfo* 2019; 21: 10–14.
11. Gholamzadeh M, Abtahi H and Safdari R. Machine learning-based techniques to improve lung transplantation outcomes and complications: a systematic review. *BMC Med Res Methodol* 2022; 22: 331.
12. Nosoudi N, Zadeh A, White H, et al. The predictive value of serum bilirubin in the post-liver transplant de novo malignancy: a data mining approach. *Healthc Anal* 2022; 2: 100094.
13. Van Trigt P, Davis RD, Shaeffer GS, et al. Survival benefits of heart and lung transplantation. *Ann Surg* 1996; 223: 576.
14. Hsich EM, Thuita L, McNamara DM, et al. Variables of importance in the scientific registry of transplant recipients database predictive of heart transplant waitlist mortality. *Am J Transplant* 2019; 19: 2067–2076.
15. Davis AE, Mehrotra S, Friedewald JJ, et al. Improving geographic equity in kidney transplantation using alternative kidney sharing and optimization modeling. *Med Decis Making* 2015; 35: 797–807.
16. Ayllón MD, Ciria R, Cruz-Ramírez M, et al. Validation of artificial neural networks as a methodology for donor-recipient matching for liver transplantation. *Liver Transplant* 2018; 24: 192–203.
17. Medved D, Nugues P and Nilsson J. *Simulating the outcome of heart allocation policies using deep neural networks*. In: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2018. IEEE.
18. Weismüller TJ, Prokein J, Becker T, et al. Prediction of survival after liver transplantation by pre-transplant parameters. *Scand J Gastroenterol* 2008; 43: 736–746.
19. Raji C and Chandra SV. Graft survival prediction in liver transplantation using artificial neural network models. *J Comput Sci* 2016; 16: 72–78.
20. Kazemi A, Kazemi K, Sami A, et al. Identifying factors that affect patient survival after orthotopic liver transplant using machine-learning techniques. *Exp Clin Transplant* 2019; 17: 775–783.
21. Kantidakis G, Putter H, Lancia C, et al. Survival prediction models since liver transplantation-comparisons between cox models and machine learning techniques. *BMC Med Res Methodol* 2020; 20: 1–14.
22. Liu C-L, Soong R-S, Lee W-C, et al. Predicting short-term survival after liver transplantation using machine learning. *Sci Rep* 2020; 10: 1–10.
23. Lau L, Kankanige Y, Rubinstein B, et al. Machine-learning algorithms predict graft failure after liver transplantation. *Transplantation* 2017; 101: e125.
24. Amini M, Bagheri A and Delen D. An explanatory analytics model for identifying factors indicative of long-versus short-term survival after lung transplantation. *Decis Anal J* 2022; 3: 100058.
25. Medved D, Ohlsson M, Höglund P, et al. Improving prediction of heart transplantation outcome using deep learning techniques. *Sci Rep* 2018; 8: 3613.
26. Zadeh A, Broach C, Nosoudi N, et al. Building analytical models for predicting de novo malignancy in pancreas transplant patients: a machine learning approach. *Expert Syst Appl* 2024; 237: 121584.
27. Zadeh AH and Sharda R. Modeling brand post popularity dynamics in online social networks. *Decis Support Syst* 2014; 65: 59–68.
28. Hassan Zadeh A, Zolbanin HM, Sharda R, et al. Social media for nowcasting flu activity: spatio-temporal big data analysis. *Inf Syst Front* 2019; 21: 743–760.
29. Hosmer Jr DW, Lemeshow S and Sturdivant RX. *Applied logistic regression*. 3rd ed. New Jersey: John Wiley & Sons, 2013.

30. Shreve J, Schneider H and Soysal O. A methodology for comparing classification methods through the assessment of model stability and validity in variable selection. *Decis Support Syst* 2011; 52: 247–257.
31. Miller A. *Subset selection in regression*. 2nd ed. London: Chapman and Hall/CRC, 2002.
32. Efron B, Hastie T, Johnstone I, et al. Least angle regression. 2004.
33. Hastie T, Tibshirani R, Friedman JH, et al. *The elements of statistical learning: data mining, inference, and prediction*. New York: Springer, 2009.
34. Breiman L**., Friedman JH, Olshen RA, Stone CJ.** *Classification and regression trees*. New York: Chapman & Hall, 2017.
35. Fuller CM, Biros DP and Delen D. An investigation of data and text mining methods for real world deception detection. *Expert Syst Appl* 2011; 38: 8392–8398.
36. Brown I and Mues C. An experimental comparison of classification algorithms for imbalanced credit scoring data sets. *Expert Syst Appl* 2012; 39: 3446–3453.
37. Emanet N, Öz HR, Bayram N, et al. A comparative analysis of machine learning methods for classification type decision problems in healthcare. *Decis Anal* 2014; 1: 1–20.
38. Friedman JH. Greedy function approximation: a gradient boosting machine. *Ann Statist* 2001; 1: 1189–1232.
39. Friedman JH. Stochastic gradient boosting. *Comput Statist Data Anal* 2002; 38: 367–378.
40. Zhang Z, Zhao Y, Canes A, et al. Predictive analytics with gradient boosting in clinical medicine. *Ann Transl Med* 2019; 7: 152.
41. Zadeh AH, Alsabi Q, Ramirez-Vick JE, et al. Characterizing basal-like triple negative breast cancer using gene expression analysis: a data mining approach. *Expert Syst Appl* 2020; 148: 113253.
42. Breiman L. Random forests. *Mach Learn* 2001; 45: 5–32.
43. Zadeh A, Combs K, Burkey B, et al. Pet analytics: predicting adoption speed of pets from their online profiles. *Expert Syst Appl* 2022; 204: 117596.
44. Bishop CM. *Neural networks for pattern recognition*. New York: Oxford university press, 1995.
45. Quinlan JR. Induction of decision trees. *Mach Learn* 1986; 1: 81–106.
46. Huang C-L, Liao H-C and Chen M-C. Prediction model building and feature selection with support vector machines in breast cancer diagnosis. *Expert Syst Appl* 2008; 34: 578–587.
47. Park BG, Park Y, Joo DJ, et al. Clinical significance of donor-specific anti-HLA-DR51/52/53 antibodies for antibody-mediated rejection in kidney transplant recipients. *Korean J Transplant* 2019; 33: 47.
48. Pati S, Irfan W, Jameel A, et al. Obesity and cancer: a current overview of epidemiology, pathogenesis, outcomes, and management. *Cancers* 2023; 15: 485.
49. Iyengar NM, Hudis CA and Dannenberg AJ. Obesity and inflammation: new insights into breast cancer development and progression. *Am Soc Clin Oncol Educ Book* 2013; 33: 46–51.
50. Steininger C. Clinical relevance of cytomegalovirus infection in patients with disorders of the immune system. *Clin Microbiol Infect* 2007; 13: 953–963.
51. Kelly MG, Winkler SS, Lentz SS, et al. Serum calcium and serum albumin are biomarkers that can discriminate malignant from benign pelvic masses. *Cancer Epidemiol Biomarkers Prev* 2015; 24: 1593–1598.
52. Yang Z, Zheng Y, Wu Z, et al. Association between pre-diagnostic serum albumin and cancer risk: results from a prospective population-based study. *Cancer Med* 2021; 10: 4054–4065.
53. Benazzo A, Schwarz S, Frommlet F, et al. Twenty-year experience with extracorporeal life support as bridge to lung transplantation. *J Thor Cardiovasc Surg* 2019; 157: 2515–2525.e2510.
54. Engels EA, Pfeiffer RM, Fraumeni JF, et al. Spectrum of cancer risk among US solid organ transplant recipients. *Jama* 2011; 306: 1891–1901.
55. Nijland ML, Kersten MJ, Pals ST, et al. Epstein-barr virus–positive posttransplant lymphoproliferative disease after solid organ transplantation: Pathogenesis, clinical manifestations, diagnosis, and management. *Transplant Direct* 2016; 2: e48.
56. Hughes R. Patient safety and quality: An evidence-based handbook for nurses. 2008.
57. Hansen R, Seifeldin R and Noe L. Medication adherence in chronic disease: issues in posttransplant immunosuppression. *Transplant Proc* 2007; 39: 1287–1300.
58. Makinodan T, Santos GW and Quinn R. Immunosuppressive drugs. *Pharmacol Rev* 1970; 22: 189–247.
59. Shtraichman O and Ahya VN. Malignancy after lung transplantation. *Ann Transl Med* 2020; 8: 416.
60. White MC, Holman DM, Boehm JE, et al. Age and cancer risk: a potentially modifiable relationship. *Am J Prevent Med* 2014; 46: S7–S15.
61. Alzayer MA. *The influence of deceased kidney donor blood urea nitrogen (BUN) level on graft and patient survival time*. New Jersey: Rutgers University-School of Health Professions, 2017.
62. Sun Y, Li J, Qu Z, et al. Causal associations between serum urea and cancer: a Mendelian randomization study. *Genes* 2021; 12: 498.
63. Wang C, Sun H and Liu J. BUN Level is associated with cancer prevalence. *Eur J Med Res* 2023; 28: 1–9.

64. Lingjun Z, Kesheng W and Xingguang L. Use of diplotypes–matched haplotype pairs from homologous chromosomes–in gene-disease association studies. *Shanghai Arch Psychiatry* 2014; 26: 165.
65. Mosaad Y. Clinical role of human leukocyte antigen in health and disease. *Scand J Immunol* 2015; 82: 283–306.
66. Nakamura T, Shirouzu T, Nakata K, et al. The role of major histocompatibility complex in organ transplantation-donor specific anti-major histocompatibility complex antibodies analysis goes to the next stage. *Int J Mol Sci* 2019; 20: 4544.
67. Shiina T, Hosomichi K, Inoko H, et al. The HLA genomic loci map: expression, interaction, diversity and disease. *J Hum Genet* 2009; 54: 15–39.
68. Chen Y, Cicciarelli J, Pravica V, et al. Long-range linkage on chromosome 6p of VEGF, FKBP5, HLA and TNF alleles associated with transplant rejection. *Mol Immunol* 2009; 47: 96–100.
69. Ferrera A, Olivo A, Alaez C, et al. HLA DQA1 and DQB1 loci in Honduran women with cervical dysplasia and invasive cervical carcinoma and their relationship to human papillomavirus infection. *Hum Biol* 1999; 1: 367–379.
70. Sabbatino F, Liguori L, Polcaro G, et al. Role of human leukocyte antigen system as a predictive biomarker for checkpoint-based immunotherapy in cancer patients. *Int J Mol Sci* 2020; 21: 7295.